

3.3 Training and Inference Strategy

Failure-aware value targets. We denote successful demonstrations by $\mathcal{D}_s$ and failure rollouts by $\mathcal{D}_f$. Each episode is annotated with its final outcome and, for failed episodes, the failure onset. We instantiate Eq. (3) as an action-conditioned progress target:

$$v_t(a_{t:t+H}^{\text{gt}}) = \begin{cases} p_{t+H}, & \text{if success,} \\ \text{clip}(p_{t+H} - \lambda_{\text{fail}} \mathbf{1}_{\text{fail}}(t+H), 0, 1), & \text{if fail,} \end{cases} \quad (6)$$

where $\mathbf{1}_{\text{fail}}(t+H)$ indicates whether failure happened by executing $a_{t:t+H}^{\text{gt}}$. In our experiments, we use a uniform progress reward for simplicity, so Eq. (3) reduces to $p_t = t/T$ for an episode of length T . This target preserves temporal progress while lowering the score assigned to action-conditioned futures that enter failure. Thus, under a failed action sequence, the model is trained to predict both the observed failed future and a lower action-conditioned progress target.

Joint denoising losses. We train all predicted modalities with the same flow-matching denoising objective [49]. For a target modality $x \in \{a, v, I\}$, let z_0^x be the clean target token and $z_1^x \sim \mathcal{N}(0, I)$ be Gaussian noise. With interpolation $z_\tau^x = (1 - \tau)z_0^x + \tau z_1^x$, the modality loss is

$$\mathcal{L}_x = \mathbb{E}_{z_0^x, z_1^x, \tau} \left[\|u_\theta^x(z_\tau^x, \tau; z) - (z_1^x - z_0^x)\|_2^2 \right], \quad (7)$$

where u_θ^x is the predicted velocity and z denotes the full token sequence in Eq. (5). Success demonstrations supervise action, value, and future prediction:

$$\mathcal{L}_{\mathcal{D}_s} = w_a \mathcal{L}_a + w_v \mathcal{L}_v + w_I \mathcal{L}_I. \quad (8)$$

For failure rollouts, we remove the action imitation term while retaining value and future-video supervision:

$$\mathcal{L}_{\mathcal{D}_f} = w_v \mathcal{L}_v + w_I \mathcal{L}_I. \quad (9)$$

Thus, failure data teaches the model the consequences and lower action-conditioned progress of failed actions without making those actions policy targets.

Two-stage inference. Inference runs the same transformer in two stages. Stage 1 denoises $[P_{\text{state}}, P_{\text{ref}}, A_{\text{noisy}}]$ for K_{denoise} flow-Euler steps and returns a clean action chunk $\hat{a}_{t:t+H}$ (in Figure 3). If only an action is required, FACT stops here and skips world prediction. When candidate scoring or consequence prediction is requested, Stage 2 places $\hat{a}_{t:t+H}$ into the clean action-conditioning slot and denoises value and, optionally, future-video latents. Since the prefix P is shared across candidates, prefix key-value caching is used in action-only inference.

Optional candidate scoring. Candidate ranking [50] is meaningful in FACT because the progress predictor is trained on action-conditioned failed outcomes. With success-only training, progress prediction is calibrated mainly on the expert action manifold and may assign overly optimistic scores to poor actions. We therefore treat value-guided selection as an optional deployment interface and diagnostic of failure-aware consequence learning, rather than an independent source of supervision. For value-guided inference, Stage 1 samples N action candidates $\{a^{(k)}\}_{k=1}^N$ in parallel. Stage 2 predicts an action-conditioned progress score $\hat{v}^{(k)} = V_\theta(o_t, \ell, a^{(k)})$ for each candidate, and the executed action is selected as

$$a^* = \arg \max_{k \in \{1, \dots, N\}} \hat{v}^{(k)} = \arg \max_{a \in a^{(1:N)}} V_\theta(o_t, \ell, a). \quad (10)$$

This selection rule uses the value head to score the future implied by each candidate action, without training a separate critic.

Implementation details. We initialize the model weights from WAN2.2-5B [51], which serves as the video diffusion backbone, and train with AdamW [52]. The learning rates are 2×10^{-4} for the action FFN and 2×10^{-5} for the WAN backbone; loss weights are $w_a = 20$ and $w_v = w_I = 1$. The action chunk length is $H = 48$ and $\lambda_{\text{fail}} = 1$. Future-video supervision uses the current frame plus four future offsets, corresponding to $[0, H/4, H/2, 3H/4, H]$. Unless otherwise specified, inference uses 20 flow-Euler denoising steps. Appendix A summarizes the failure-aware co-training loop, rollout failure collection, and two-stage inference procedure in pseudocode.